

GenAI Under Attack: Evaluating Enterprise Security Gaps Against Emerging AI Threat Vectors

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The integration of generative AI (GenAI) has introduced new attack vectors that existing cybersecurity frameworks were not designed to adequately address. This project draws on GenAI vulnerability reports from 2022–2025 and early 2026, cross-referenced against OWASP LLM Top 10 and MITRE ATLAS classifications, to evaluate which attack vectors current security controls leave most exposed and what mitigation strategies most effectively close that gap.

The OWASP LLM Top 10 provides GenAI application-layer risk classifications, while MITRE ATLAS maps adversarial tactics and techniques specific to AI and ML systems. The rapid maturation of MITRE ATLAS, from 36 techniques across 12 tactics in 2021 to 97 techniques across 16 tactics by March 2026, reflects the accelerating pace at which the AI attack surface is being formally documented.

The defensive tooling ecosystem has matured considerably since 2022, producing purpose-built solutions across pre-deployment evaluation, adversarial red teaming, input filtering, output validation, and continuous monitoring. Pre-deployment adversarial red teaming emerges as the highest-leverage intervention, addressing attack vectors that neither static code analysis nor runtime monitoring can detect. However, no single tool provides adequate protection. Training data poisoning remains underserved, multimodal attack vectors are outpacing text-only input defenses, and the architectural root cause of prompt injection between instructions and data cannot be addressed by any currently available tool.

Attack vector priorities are constantly evolving. While prompt injection, model inversion, and data poisoning defined the 2024 - 2025 threat landscape, the current threat surface as of early 2026 has shifted toward LLM jailbreaking, which overlaps substantially with prompt injection, and the rapidly maturing attack surface introduced by Model Context Protocol (MCP) integrations. This research finds that as AI becomes embedded in critical systems, an accurate and current taxonomy of attack vectors is foundational to effective cybersecurity posture.